

Project 4: Auditing Three Pretrained CelebA Classifiers

Group 4

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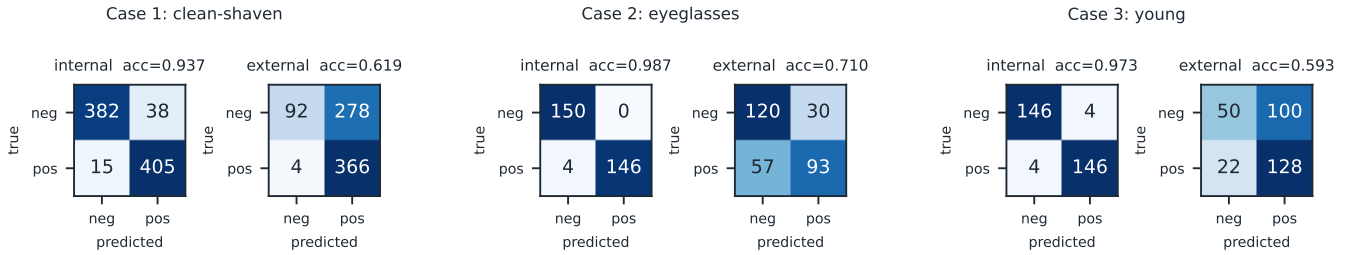


Figure 1. Confusion matrices on both splits of each case, counts shown. The three failures have different shapes: case 1 floods false positives (278 externally, 38 internally), case 2 misses 57 positives while making only 30 false positives, and case 3 errs on both sides while staying confident.

Three ResNet-18 classifiers, each built by a different team for a different CelebA attribute, passed internal testing and then failed in deployment. Each was reloaded unchanged and evaluated on its supplied splits, with no retraining anywhere. Every case ran the same battery: the image statistics of all four class folders before any prediction was made, then confusion matrices, precision, recall and F1, ROC and AUC for discrimination, a ten-bin reliability diagram with expected calibration error, the accuracy of the most confident predictions, and class activation maps. Each case then ran one further experiment, designed so that a negative result would have falsified its diagnosis. Aggregate accuracy cannot separate the three faults, since all three land in a similar band externally (Table 1); the other axes are what distinguish them.

1. Case 1: clean-shaven, studio to mobile

Diagnosis: texture dependency, exposed by a loss of image detail between studio and mobile capture. The classifier reads “clean-shaven” from fine stubble texture on the chin and jaw. The mobile images carry a fraction of the detail the studio images do, so that texture is gone and a face that is not clean-shaven no longer looks different from one that is.

Experiments and evidence. At identical pixel dimensions and file format, the field images have a Laplacian variance 9.9 times lower than the studio ones, 0.00109 against 0.01077, and are 0.262 brighter. The same measurement on cases 2 and 3 gives ratios of 1.2 and 0.9, so the gap belongs to this case rather than to the way it is measured. The errors are one-sided in the direction that texture loss predicts: false positives rise from 38 to 278 while recall *rises* to 0.989.

The decisive test blurs the studio images and changes nothing else, so the subjects and the labels are identical throughout. Accuracy falls monotonically with blur and passes through the field result: at a Gaussian sigma of 4.0 pixels it reaches 0.620 against the field’s 0.619, and arrives with the same shape, 310 false positives against 278 (Figure 2, left). The brightness shift applied on its own reaches only 0.918, so exposure differs between the splits but does not cause the failure. Gaussian blur is a stand-in for whatever the mobile pipeline does rather than a claim about it; what the sweep establishes is that

removing detail is sufficient to produce the failure, so a field set of different or harder subjects is not needed to explain it. Saliency agrees: splitting each activation map into three bands of the face shows 0.542 of the attention on the mouth and jaw when the model is right, falling to 0.327 when it is wrong while the forehead and hair share rises from 0.042 to 0.242. Once the texture is gone the region that should decide the prediction holds nothing to decide it with.

Recommendations. Gate on sharpness at inference. Laplacian variance is one cheap statistic per frame, and a frame below the studio range should prompt a retake rather than be scored. Match the pre-processing to the capture path so the model is not asked for detail the camera never recorded. The real fix is to extend the training data with mobile-quality captures and validate on a held-out mobile sample, because a detector for a texture cue can only be trusted where the texture survives.

2. Case 2: eyeglasses at a retail kiosk

Diagnosis: a spurious correlation between eyeglasses and the subject’s sex, learned because the internal set contains only two of the four possible combinations of the two.

Experiments and evidence. Displaying all four class folders (Figure 2, centre) shows the internal set is built from women without eyeglasses and men wearing them, while the external set holds exactly the two combinations that never appear internally. Eyeglasses and sex are therefore perfectly confounded in everything the model was validated against, and a classifier responding only to sex would have reached close to the reported internal accuracy of 0.987 without representing eyeglasses at all.

Stratifying the same predictions by combination separates the two explanations that aggregate accuracy conflates. It reaches 0.987 accuracy on the two combinations present internally and 0.710 on the two absent from it, an error rate 22 times higher. The errors fall on both sides exactly as a sex cue predicts, missing eyeglasses on 57 of the 150 women wearing them and reporting eyeglasses on 30 men who are not. A model that had merely degraded would err evenly across subjects; this one errs according to whether the subject matches the pairing it was trained under, which is precisely the reported symp-



Figure 2: The experiment that identifies each fault. **Left, case 1:** field images beside studio ones, and the accuracy of the studio set as blur is swept across it; blur alone reaches the field result, the brightness shift does not. **Centre, case 2:** the four class folders, and accuracy split by attribute combination, with the two combinations the internal set contained outlined. **Right, case 3:** calibration measured on a held-out half of the field set, and accuracy against the fraction published when predictions are ordered by confidence.

tom of working for some users and not others on images of equal quality.

Saliency measures the same thing directly. Splitting each activation map into three horizontal bands of the face shows that on predictions the model gets right, 0.552 of its attention falls on the eyes and nose, where eyeglasses are. On the ones it gets wrong that share drops to 0.325 and the largest share, 0.498, moves to the mouth and jaw. The model looks at the eyeglasses when it succeeds and elsewhere when it fails.

Recommendations. Recalibration cannot help, because the fault is in the representation rather than in the confidence attached to it. Rebuild the validation set so eyeglasses and sex vary independently, which means collecting the two missing combinations, and report performance per combination rather than in aggregate, since an aggregate over an unbalanced set concealed this for the whole lab phase. Until then the kiosk should confirm with the user rather than act on the prediction, which costs one tap and avoids a recommendation resting on a demographic inference nobody specified.

3. Case 3: “appears young” with an auto-publish bypass

Diagnosis: overconfidence under distribution shift. The model does not transfer to the deployed population and its confidence does not reveal it; validating on an unrepresentative internal set is the cause. The internal negatives are clearly elderly faces while the field negatives are middle-aged, so the lab phase measured a much easier problem than deployment poses.

Experiments and evidence. Internally the model reaches AUC 0.9960 and is reasonably calibrated at ECE 0.036, so the bypass looked safe. Externally AUC falls to

Case	Split	Acc	P	R	F1	AUC	ECE
1 clean-shaven	internal	0.937	0.914	0.964	0.939	0.9845	0.013
1 clean-shaven	external	0.619	0.568	0.989	0.722	0.8459	0.250
2 eyeglasses	internal	0.987	1.000	0.973	0.986	0.9997	0.013
2 eyeglasses	external	0.710	0.756	0.620	0.681	0.8125	0.152
3 young	internal	0.973	0.973	0.973	0.973	0.9960	0.036
3 young	external	0.593	0.561	0.853	0.677	0.5640	0.296

Table 1: All three cases fall into a similar accuracy band externally. AUC, which measures discrimination, and ECE, which measures whether the confidence can be believed, are what separate them: case 1 keeps most of its ranking, case 3 loses nearly all of it, and only case 3 is badly calibrated internally as well.

0.5640, close to the 0.5 of a coin toss: the scores no longer rank the classes. The softmax nevertheless keeps emitting extreme probabilities, so ECE rises to 0.296. The consequence for the bypass is direct. Of the field predictions issued at 0.90 confidence or above, 54.3 percent of the split, only 0.558 are correct. Raising the gate to 0.99 leaves 20.0 percent published at 0.517, so tightening it does not help: the confidence that selects them is no longer informative about whether they are right. Both thresholds are reported because the assessment page and the supplied notebook state different ones, and the conclusion is the same at either.

Fitting a temperature on half the field set and measuring on the other half confirms the mechanism. Expected calibration error falls from 0.326 to 0.070 while AUC stays at 0.544, unchanged, because dividing a logit by a constant is monotone and cannot reorder predictions. The fitted temperature is 12.16, and after scaling the most confident field image reaches only 0.687, so a gate at either threshold would publish 0 of the 150 held-out images. The risk-coverage curve states the ceiling directly: the best accuracy reachable at any coverage above two percent of the field split is 0.800 (Figure 2, right).

Recommendations. Switch the bypass off rather than re-tune its threshold. No threshold publishes at 0.90 accuracy on this population, so any gate setting trades volume against errors instead of avoiding them. Recalibrate, but treat it as instrumentation rather than a repair: a calibrated score is what lets monitoring detect this condition, and the fact that almost nothing then clears the gate is the honest reading, not a defect of the calibration. Rebuild the internal set so its negatives span the ages the system actually meets, and monitor expected calibration error and high-confidence accuracy continuously, because both looked healthy in the lab and neither stayed so.

4. Summary

The three models fail at similar accuracy for different reasons, which is why the remedies differ. Case 1 is a property of the inputs, case 2 of the training distribution, and case 3 of the validation design. In all three the aggregate metric that passed the lab phase is the same metric that hid the fault, which is the general lesson: an audit has to compare distributions and stratify results rather than summarise them.